

Ahmed H. Mahmoud

Curriculum Vitae

<https://ahdhn.github.io>
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RESEARCH EXPERIENCE

Massachusetts Institute of Technology, CSAIL

September 2024 – present

Postdoctoral Associate

- Advisor: Professor Justin Solomon & Professor Jonathan Ragan-Kelley

Autodesk Research, Toronto, Canada

November 2020 – May 2024

Senior Research Scientist

University of California, Davis

March 2016 – October 2020

Graduate Student Researcher

Autodesk Research, Toronto, Canada

June–December 2019, July – November 2020

Intern, Numerical Analysis Research

Shenzhen University, China

June – September 2018

Research intern at the Visual Computing Research Center

EDUCATION

University of California, Davis

June 2024

Ph.D. in Electrical and Computer Engineering

- Advisor: Professor John D. Owens
- Dissertation: Unstructured Geometric Data Processing on the GPU: Data Structures & Programming Models

University of California, Davis

September 2020

M.Sc. in Electrical and Computer Engineering

Alexandria University, Egypt

May 2013

B.S. in Marine Engineering and Naval Architecture

- Very good with honors—ranked first

FUNDING

[1] OAC Core: OAC Core Projects: GPU Geometric Data Processing

National Science Foundation (Award # OAC-2403239)

PI. Jonathan Ragan-Kelley, co-PI. Justin Solomon

Amount: **\$600,000**

July 1, 2024–June 30, 2027

Role: The primary author of the technical description

[2] Efficient GPU Sparse Automatic Differentiation for Scientific Computing

MIT Generative AI Impact Consortium (MGAIC)

PI. Justin Solomon, co-PI. Jonathan Ragan-Kelley

Amount: **\$150,000**

June 1, 2025–May 31, 2026

Role: The primary author of the technical description

PUBLICATIONS

- [1] *Locality-Aware Automatic Differentiation on the GPU for Mesh-Based Computations*
Ahmed H. Mahmoud, Rahul Goel, Jonathan Ragan-Kelley, and Justin Solomon.
ACM Transactions on Graphics (SIGGRAPH 2026).
- [2] *Fast Sparse Matrix Permutation for Mesh-Based Direct Solvers*
Behrooz Zarebavani[†], **Ahmed H. Mahmoud**[†], Ana Dodik, Changcheng Yuan, Serban D. Porumbescu, John D. Owens, Maryam Mehri Dehnavi, and Justin Solomon.
SIGGRAPH 2026.
[†] joint first author
- [3] *iskra: A System for Inverse Geometry Processing*
Ana Dodik, **Ahmed H. Mahmoud**, and Justin Solomon.
ACM Transactions on Graphics (SIGGRAPH 2026).
- [4] *Low-Rank Adaptation of Neural Fields*
Anh Truong, **Ahmed H. Mahmoud**, Mina Konaković Luković, and Justin Solomon.
SIGGRAPH Asia 2025.
- [5] *Dynamic Mesh Processing on the GPU*
Ahmed H. Mahmoud, Serban D. Porumbescu, and John D. Owens.
ACM Transactions on Graphics (SIGGRAPH 2025).
- [6] *Disaggregated Design for GPU-Based Volumetric Data Structures*
Massimiliano Meneghin and **Ahmed H. Mahmoud**.
European Conference on Parallel and Distributed Computing (EuroPar 2025)
- [7] *Optimized GPU implementation of grid refinement in lattice Boltzmann method*
Ahmed H. Mahmoud, Hesam Salehipour, and Massimiliano Meneghin
International Parallel and Distributed Processing Symposium (IPDPS 2024)
Open Source Contribution Award
- [8] *Neon: A Multi-GPU Programming Model for Grid-based Computations*
Massimiliano Meneghin[†], **Ahmed H. Mahmoud**[†], Pradeep Kumar Jayaraman, and Nigel J. W. Morris.
International Parallel and Distributed Processing Symposium (IPDPS 2022)
[†] joint first author
- [9] *RXMesh: A GPU Mesh Data Structure*
Ahmed H. Mahmoud, Serban D. Porumbescu, and John D. Owens
ACM Transactions on Graphics (SIGGRAPH 2021)
- [10] *VoroCrust: Voronoi Meshing Without Clipping*
Ahmed Abdelkader, Chandrajit L. Bajaj, Mohamed S. Ebeida, **Ahmed H. Mahmoud**, Scott A. Mitchell, John D. Owens and Ahmad A. Rushdi
ACM Transactions on Graphics (SIGGRAPH 2020)
- [11] *Sampling Conditions for Conforming Voronoi Meshing by the VoroCrust Algorithm*
Ahmed Abdelkader, Chandrajit L. Bajaj, Mohamed S. Ebeida, **Ahmed H. Mahmoud**, Scott A. Mitchell, John D. Owens and Ahmad A. Rushdi
International Symposium on Computational Geometry (SoCG 2018)
- [12] *A Constrained Resampling Strategy for Mesh Improvement*
Ahmed Abdelkader[†], **Ahmed H. Mahmoud**[†], Ahmad A. Rushdi, Scott A. Mitchell, John D. Owens, and Mohamed S. Ebeida
Computer Graphics Forum (presented at ACM/Eurographics Symposium on Geometry Processing SGP 2017)
[†] joint first author
- [13] *All-Quad Meshing without Cleanup*
Ahmad A. Rushdi, Scott A. Mitchell, **Ahmed H. Mahmoud**, Chandrajit L. Bajaj, and Mohamed S. Ebeida
Computer-Aided Design (CAD 2017)

- [14] *Disk Density Tuning of a Maximal Random Packing*
Mohamed S. Ebeida, Ahmad A. Rushdi, Muhammad A. Awad, **Ahmed H. Mahmoud**, Dongming Yan, Shawn English, John D. Owens, Chandrajit L. Bajaj, and Scott A. Mitchell
Computer Graphics Forum (presented at ACM/Eurographics Symposium on Geometry Processing SGP 2016)
- [15] *Exercises in High-Dimensional Sampling: Maximal Poisson-disk Sampling and k-d Darts*
Mohamed S. Ebeida, Scott A. Mitchell, Anjul Patney, Andrew A. Davidson, Stanley Tzeng, Muhammad A. Awad, **Ahmed H. Mahmoud**, and John D. Owens
Book chapter in Topological and Statistical Methods for Complex Data: Tackling Large-Scale, High-Dimensional, and Multi-variate Data Spaces (2014)
- [16] *Delaunay Quadrangulation by Two-coloring Vertices*
Scott A. Mitchell, Mohammed A. Mohammed, **Ahmed H. Mahmoud** and Mohamed S. Ebeida
International Meshing Roundtable (IMR 2014)
- [17] *Improving Spatial Coverage while Preserving the Blue Noise of Point Sets*
Mohamed S. Ebeida, Muhammad A. Awad, Xiaoyin Ge, **Ahmed H. Mahmoud**, Scott A. Mitchell, Patrick M. Knupp, and Li-Yi Wei
SIAM Conference on Geometric and Physical Modeling (SIAM GD/SPM13)
- [18] *Sifted Disks*
Mohamed S. Ebeida, **Ahmed H. Mahmoud**, Muhammad A. Awad, Mohammed A. Mohammed, Scott A. Mitchell, Alex Rand, and John D. Owens
Eurographics 2013.

AWARDS

- [1] Symposium on Geometry Processing (SGP) 2026 Software Award
Ahmed H. Mahmoud, Serban D. Porumbescu, and John D. Owens
SGP Software Award recognizes “the authors of an open-source software project that has greatly influenced the field.”
- [2] IPDPS 2024 Open Source Contribution Award
Ahmed H. Mahmoud, Hesam Salehipour, and Massimiliano Meneghin
For the Optimized GPU Implementation of Grid Refinement in the Lattice Boltzmann Method

PATENTS

- [1] *Optimized GPU Implementation of Grid Refinement in the Lattice Boltzmann Method*
Ahmed H. Mahmoud, Hesam Salehipour, and Massimiliano Meneghin
Filed on January 29, 2025 by Autodesk, Inc. [US20250307504A1](#)

SELECTED TALKS

Accelerating Geometric Optimization on GPUs

Summer Geometry Initiative (August 2026 - Virtual)

Locality-Aware Automatic Differentiation on the GPU for Mesh-Based Computations

SIGGRAPH 2026 (August 2026 - CA, USA)

Fast Sparse Matrix Permutation for Mesh-Based Direct Solvers

SIGGRAPH 2026 (August 2026 - CA, USA)

Accelerating Irregular Computation

Queen's University (May 2026 - Kingston, Canada)
University of Toronto (April 2026 - Toronto, Canada)
University of Waterloo (March 2026 - Waterloo, Canada)
University of California, Berkeley (February 2026 - CA, USA)
University of Victoria (November 2025 - Victoria, Canada)

Dynamic Mesh Processing on the GPU

SIGGRAPH (August 2025 - Vancouver, Canada)
Highlights of Parallel Computing (July 2025 - Portland, Oregon)
Brown Visual Computing Seminar (October 2024 - Brown University)
Adobe (November 2023 - Virtual)

RXMesh: A High-performance Mesh Data Structure and Programming Model on the GPU

NVIDIA GTC (March 2022 - Virtual)

Neon: A Multi-GPU Programming Model for Grid-based Computations

NVIDIA GTC (March 2022 - Virtual)

RXMesh: A GPU Mesh Data Structure

SIGGRAPH (August 2021 - Virtual)

A Constrained Resampling Strategy for Mesh Improvement

ACM/Eurographics Symposium on Geometry Processing (July 2017 - London, UK)

MENTEES

Jinyuan Wu (Undergrad Student, MIT)

Project: Accelerated Differentiable Elastic Simulation

Behrooz Zarebavani (Ph.D. Student, University of Toronto) → SIGGRAPH paper

Project: Fast Sparse Matrix Permutation for Mesh-Based Direct Solvers

Anh Truong (Ph.D. Student, MIT) → First SIGGRAPH Asia paper

Project: Parameter-efficient Updates of Neural Fields using LoRA

Sachin Kishan (Undergrad Student, Vellore Institute of Technology) → Ph.D. Student at New York University

Project: GPU Geometric Multigrid on Triangle Mesh

Changcheng (Eric) Yuan (M.Sc. Student, UC Davis) → Ph.D. Student, Texas A&M University

Project: GPU Sparse Matrix Reordering

Brooke Dolny (Autodesk Research intern) → M.Sc. Student, University of Waterloo

Project: GPU-accelerated Lattice-Boltzmann Fluid Simulation

Gokul Adithya and Yufan Hu (Summer Geometry Initiative 2026)

Project: Inverse Geodesic Distance with Iskra

Renan Bomtempo, Adan Javier Ortiz, Daryna Hrybchuk, and Francisco Vigil (Summer Geometry Initiative 2025)

Project: Localized Remeshing for Cloth Simulation

Aniket Rajnish, Charuka Bandara, Diany Pressato, Kenia Nova, and Sachin Kishan (Summer Geometry Initiative 2024)

Project: GPU-accelerated Geometry Processing Applications

TEACHING

Differentiable Geometry Processing in Python ([YouTube](#))

2026

Symposium of Geometry Processing course

Accelerated Computing (6.S894) MIT Guest Lecturer	<i>Fall 2024, Fall 2025</i>
Control Systems I (EECS 157A) University of California, Davis Teaching Assistant	<i>Fall 2017</i>
Computer Programming (CS224) Alexandria University, Egypt Assistant Lecturer	<i>Fall 2015</i>
Ships and Machines Drawing (MR111) Alexandria University, Egypt Assistant Lecturer	<i>Fall 2015</i>
Fluid Mechanics (MR231) Alexandria University, Egypt Assistant Lecturer	<i>Fall 2015</i>
Fluid Mechanics and Hydraulic Machines (MR232) Alexandria University, Egypt Assistant Lecturer	<i>Spring 2014</i>
Marine Hydro-dynamics (OCE323) Alexandria University, Egypt Assistant Lecturer	<i>Spring 2014</i>
Theory of Machines (ME145) Alexandria University, Egypt Assistant Lecturer	<i>Spring 2014</i>
Material Technology (MR242) Alexandria University, Egypt Assistant Lecturer	<i>Fall 2013</i>
Marine Power Plants (MR352) Alexandria University, Egypt Assistant Lecturer	<i>Fall 2013</i>

ACADEMIC SERVICE

SIGGRAPH Technical Papers Committee Member	<i>2026, 2027</i>
SIGGRAPH Posters Juror	<i>2025, 2026</i>
MIT Summer Geometry Initiative Admission Committee and mentor	<i>2025, 2026</i>
New England Symposium on Graphics Organizing Committee	<i>2025, 2026</i>
ACM/Eurographics Symposium on Geometry Processing Technical Papers Committee	<i>2024, 2025</i>
MIT Summer Geometry Initiative Mentor	<i>2024</i>

High Performance Graphics	2024
International Paper Committee	
International Conference on Geometric Modeling and Processing	2023, 2024
Technical Program Committee	
ECE Peer Mentoring Program at UC Davis	2021, 2023
Mentor	
UC Davis SACNAS's Mentor Match Program	2023
Mentor	

REFeree SERVICE

ACM Transactions on Architecture and Code Optimization	2026
ACM Transactions on Parallel Computing	2026
SIGGRAPH	2024, 2025
SIGGRAPH Asia	2024
Computers & Graphics	2024
Transactions on Visualization and Computer Graphics	2023
Eurographics	2023
Computer Aided Geometric Design	2022
The SIAM International Meshing Roundtable Workshop	2022, 2023, 2024
International Meshing Roundtable	2019, 2021
Computer-Aided Design	2019

MEDIA COVERAGE

Sandia LabNews	<i>April 2020</i>
Automating complex 3D modeling (webpage , pdf)	

REFERENCES

John D. Owens

Child Family Professor of Engineering and Entrepreneurship – University of California, Davis
jowens@ece.ucdavis.edu

Justin Solomon

Associate Professor – Massachusetts Institute of Technology
jsolomon@mit.edu

Jonathan Ragan-Kelley

Associate Professor – Massachusetts Institute of Technology
jrk@mit.edu